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IS-253: Programming Language (Python)

Elchin Salahzada

Lecture 1

Agenda



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- Course Syllabus
- Course Overview
- Course Material
- Grading Policy
- Introduction to Programming Languages

Course Syllabus



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- Introduction to Programming Languages
- Operators, Expressions & I/O
- Control Structures
- Data Structures I - Lists & Tuples
- Data Structures II - Dictionaries & Sets
- Functions & Scope
- Object Oriented Programming I - Classes & objects
- Object Oriented Programming II - Inheritance & Polymorphism
- Error Handling, File I/O
- Modules, Libraries & Data Handling
- SDLC & Further Improvements



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Course Overview

- The course teaches you how to code on Python Programming Language
- This is not a course on:
 - – How to build GUI applications
 - - Data Science



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Course Material

- **"Python Crash Course"** by Eric Matthes - good for beginners
- **"Automate the Boring Stuff with Python"** by Al Sweigart - better choice for being advance
- **"Fluent Python"** by Luciano Ramalho - good for beginners



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Grading

Class Activity	10%
Midterm	30%
Lab Activity	20%
Final Exam	40%



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LAB Assignment

- LAB Assignment is used to be sure that you've understood the topic and will have no problem in exams.
- Deadlines for assignments will be set to 5-7 days from announcement.
- It is highly recommended to start working on assignments on the announcement day.
- For each missed deadline day you will get 25% penalty per day.
- Assignments are individual work, so solutions must be individual.
- You may use search engines however, using AI tools are not allowed.



Class Activity

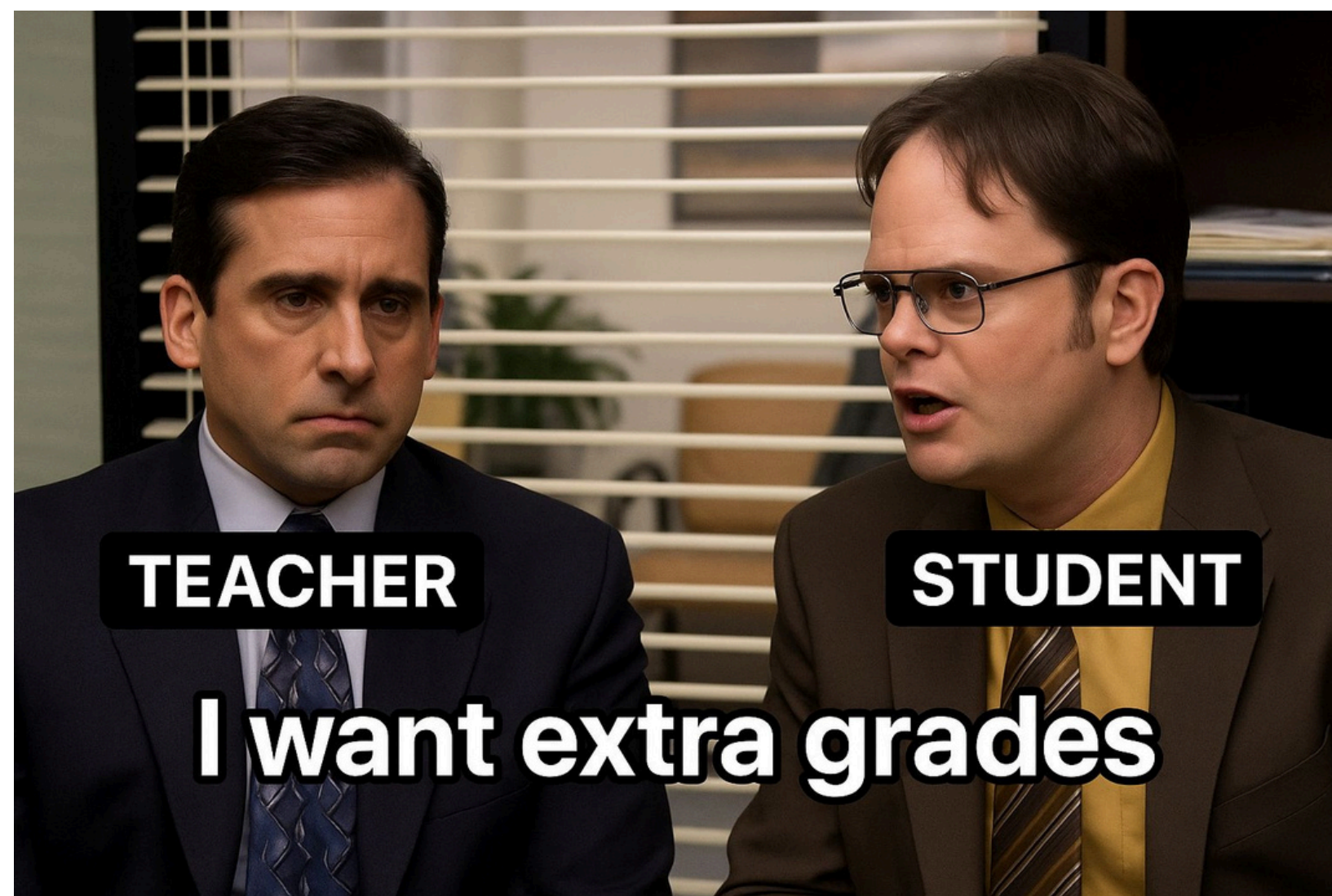
- During lab sessions, students must practice the corresponding lecture material.
- Students' class activity will be graded daily during lectures.
- Students may get better grades, if they:
 - Asking relevant questions regarding lecture
 - Answering questions
 - Any other lecture and lab related class activity
- **Misbehavior in class will reduce your daily activity points.**
- Examples of misbehavior include:
 - Using computers for non-class purposes
 - Using personal devices during lectures (phone, personal laptop)
 - Talking with classmates during lectures and labs
 - Any behavior irrelevant to class activities will not be tolerated



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Don't

- No asking for extra points, adjustments, curve, etc.
- If you need higher GPA, think about it now.





Plagiarism

- Plagiarism in any form is strictly prohibited. The following actions constitute plagiarism:
 - Copying solutions from other students (all parties involved will be held equally responsible)
 - Using AI-generated content
 - Copying material from internet sources without proper attribution
- Plagiarism is detected through specialized software (TurnitinAI Detector).
- Any confirmed violation will result in a 5-point deduction from your Class Activity grade and corresponding assignment will be graded as 0.

Introduction to Programming languages



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- What is a Programming Language
- Machine Code & Translation
- Programming Paradigms
- Low-Level vs High-Level Languages
- Compiled vs Interpreted Languages
- Introduction to Python
- Why Python is Popular
- The Python Interpreter
- Setting Up & Running Python

IS-253: Programming Language

Baku Higher Oil School : Information Security



Learning objectives

- Define what a programming language is
- Explain how computers translate instructions into machine code
- Identify different programming paradigms
- Differentiate between low-level and high-level languages
- Differentiate between compiled and interpreted languages
- Describe what Python is and its role among programming languages
- Explain why Python is widely used today
- Explain how the Python interpreter executes code
- Install Python and run a simple program

Introduction to Programming Languages



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Python



JavaScript



SQL



Java



PHP



C#



C++



TypeScript



Golang



Rust



Ruby



C



Swift



R



Introduction to PL

- A programming language is a formal set of rules and syntax used to give instructions to a computer
- It allows humans to write logic in a structured way that a machine can eventually understand
- Every language has:
 - Syntax — the grammar rules (how instructions must be written)
 - Semantics — the meaning behind those instructions
- Programming languages exist so we can express what we want a computer to do, without manually writing binary code
- Example: instead of writing raw binary, you can write `if age >= 18: print("Adult")` — a human-readable instruction that gets translated for the machine

Machine Code & Translation



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`x += 4;`

`add $s1, $s1, 4`

`0010001000110001
0000000000000000100`

@}



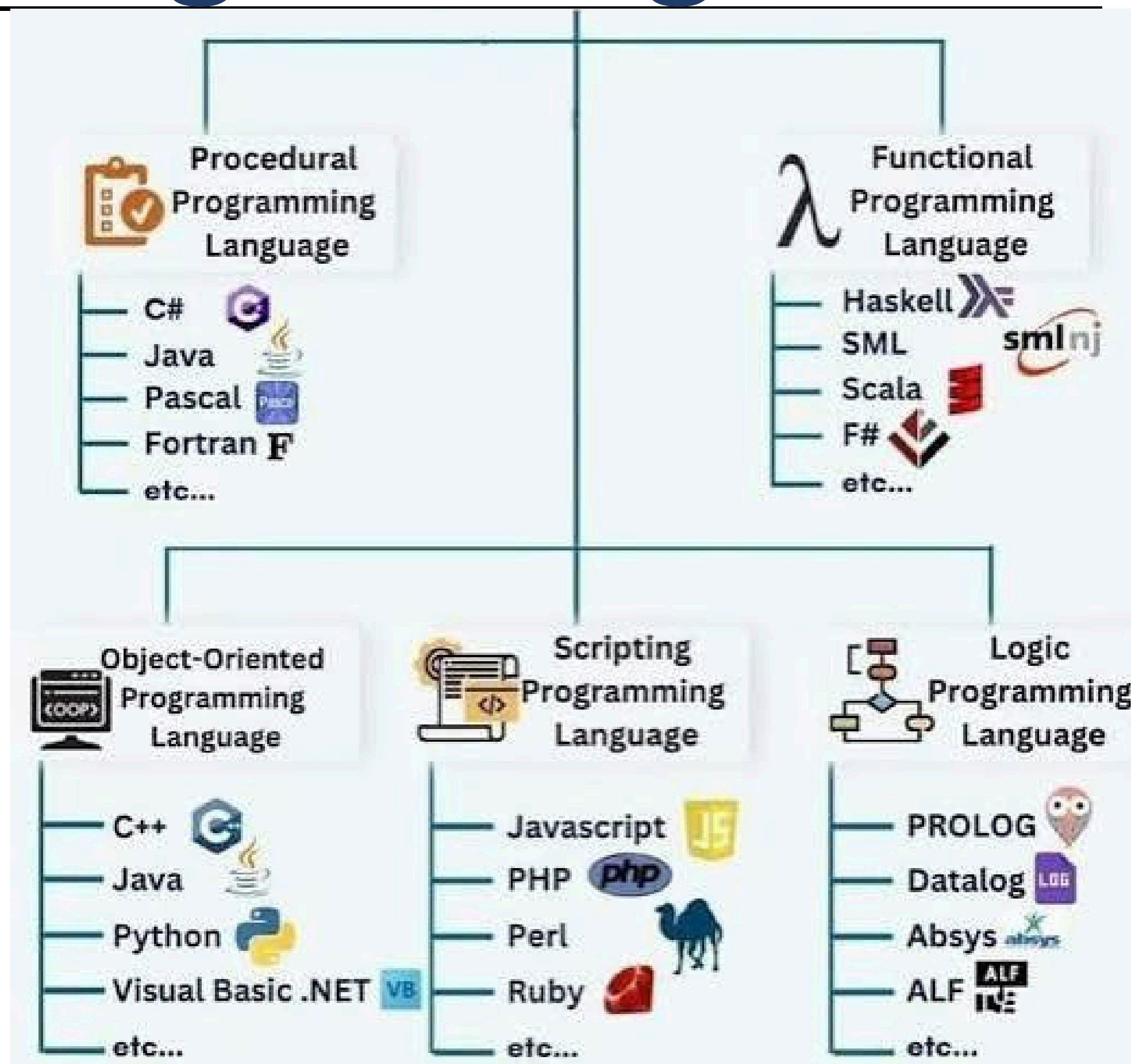
Machine Code & Translation

- Computers only understand binary — sequences of 0s and 1s known as machine code
- Every instruction, no matter what language it's written in, must eventually become machine code before the CPU can execute it
- Writing programs directly in binary would be extremely slow, complex, and error-prone for humans
- Programming languages solve this by acting as a translation layer between human logic and machine-level instructions:
- Human-readable code → Translation → Machine code → CPU executes
- This translation happens in one of two ways — compilation or interpretation — which we'll cover in the next slides

Programming Paradigms



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Programming Paradigms



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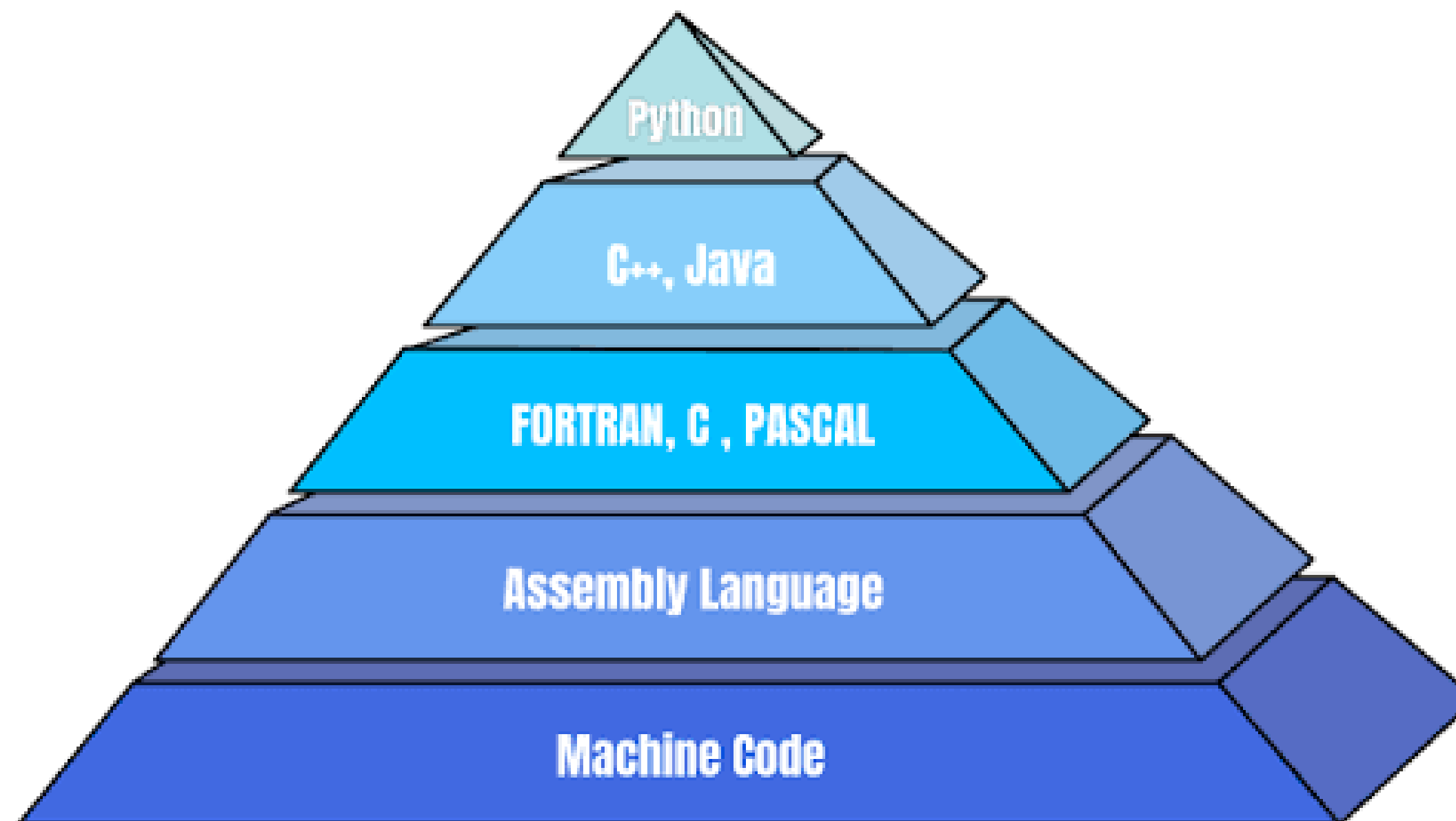
- A paradigm is a style or approach to writing and organizing code
- Different paradigms offer different ways of thinking about and structuring a solution
- Common paradigms include:
 - Procedural — code is organized as a sequence of steps/functions (e.g., C)
 - Object-Oriented (OOP) — code is organized around "objects" that bundle data and behavior together (e.g., Java, Python)
 - Functional — code is built by combining pure functions, avoiding changing state (e.g., Haskell)
 - Scripting — lightweight code often used for automation and quick tasks (e.g., Python, JavaScript)
- Python supports multiple paradigms — mainly procedural and object-oriented — giving you flexibility in how you write code

Low-Level vs High-Level



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High Level Language



Low Level Language



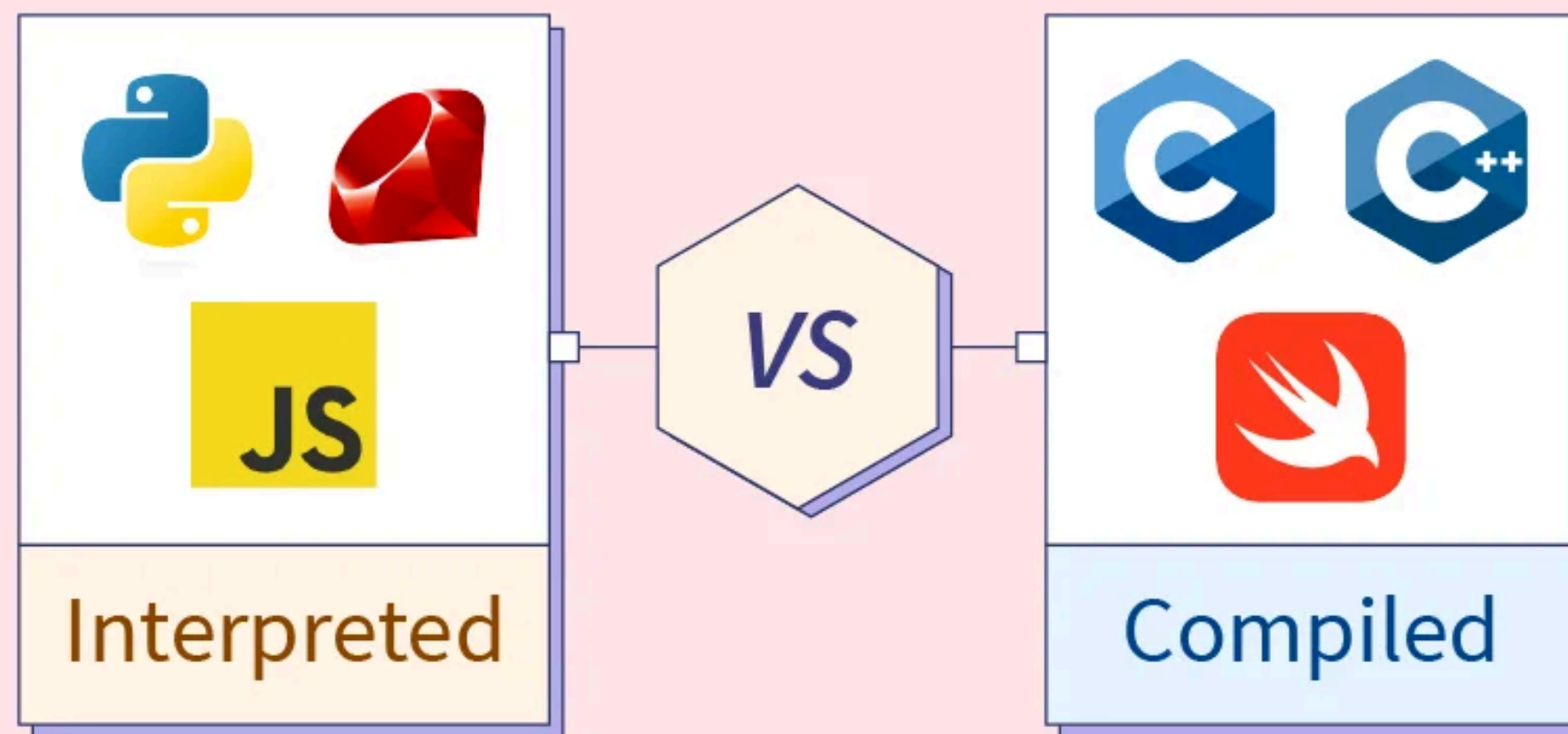
Low-Level vs High-Level

- Low-Level Languages
 - Closer to machine code, further from human language
 - Give precise control over hardware (memory, CPU registers)
 - Harder to read, write, and maintain
 - Hardware-specific — not easily portable across systems
 - Example: Assembly language
- High-Level Languages
 - Closer to human language, easier to read and write
 - Abstract away hardware details, letting programmers focus on logic
 - Portable — can run on different systems with little/no change
 - Example: Python, Java, C++
- Python is a high-level language, which is a major reason it's easy to learn and widely used

Compiled vs Interpreted



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Compiled vs Interpreted



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- Compiled Languages
 - Entire source code is translated into machine code before running, by a compiler
 - Produces a standalone executable file
 - Faster execution, but must be recompiled after every code change
 - Example: C, C++
- Interpreted Languages
 - Source code is translated and executed line-by-line, at runtime, by an interpreter
 - No separate compilation step — easier to test and debug
 - Generally slower than compiled languages, but faster to develop with
 - Example: Python, JavaScript
- Python is an interpreted language — this is why you can run code immediately without a build step

Introduction to Python



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- Python is a high-level, interpreted, general-purpose programming language
- Created by Guido van Rossum, first released in 1991
- Python was heavily inspired by and built as a successor to the ABC language, aiming to fix its limitations while keeping its simplicity
- Also influenced by languages like C, Modula-3, and SmallTalk
- Designed with a strong focus on code readability and simplicity
- Named after the British comedy show "Monty Python's Flying Circus" — not the snake
- Supports multiple paradigms: procedural and object-oriented programming
- Comes with a large standard library and an even larger ecosystem of third-party packages
- Used across many domains: web development, data science, automation, artificial intelligence, scripting, and more

Why Python is popular?



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- Simple & Readable Syntax — code resembles plain English, making it beginner-friendly
- Versatile — used in web development, data science, AI/ML, automation, scripting, game development, and more
- Large Standard Library — comes with built-in modules for many common tasks (file handling, math, networking, etc.)
- Huge Ecosystem of Third-Party Packages — libraries like NumPy, Pandas, Django, TensorFlow extend Python's capabilities
- Cross-Platform — runs on Windows, macOS, Linux without modification
- Strong Community Support — extensive documentation, tutorials, and active developer community
- High Demand in the Job Market — widely used in top tech companies and startups alike
- Great for Beginners and Experts Alike — easy to start with, yet powerful enough for complex, large-scale systems

The Python Interpreter



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- Python code (.py file) is executed by the Python Interpreter, not compiled into a standalone executable
- Execution flow:
 - Source Code (.py) → Interpreter reads line-by-line → Converts to bytecode → Executed by Python Virtual Machine (PVM)
- This is why Python is called an interpreted language — there's no separate compilation step before running
- Benefits of this approach:
- Immediate feedback — run code without waiting for compilation
- Easier debugging — errors are shown as they occur, line-by-line
- Platform independence — same code runs on different operating systems, as long as Python is installed
- Trade-off: interpreted execution is generally slower than compiled languages like C or C++
- The standard Python interpreter is called CPython — the official and most widely used implementation

Setting Up & Running Python



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- Step 1: Download Python
 - Visit the official website: python.org
 - Download the latest stable version for your operating system
- Step 2: Install Python
 - Run the installer
 - Important: check the box "Add Python to PATH" during installation (Windows) so it can be run from any terminal
- Step 3: Choose a Code Editor / IDE
 - Beginner-friendly options: VS Code, Jupyter Notebook
 - More advanced option: PyCharm
- Step 4: Write and Run Your First Program
 - Create a file named `hello.py`
 - Write the following code:
 - `print("Hello, World!")`
 - Run it in the terminal using:
 - `python hello.py`

Output



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- After running last command it should display like below:

Untitled-1.py X

C: > Users > SalahzadaE > Desktop > Untitled-1.py

```
1 print("Hello World!")
```

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE

```
/Users/SalahzadaE/Desktop/Untitled-1.py
```

```
Hello World!
```

```
PS C:\Users\SalahzadaE\AppData\Local\Programs\Microsoft VS Code>
```

Lecture Summary



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- Programming languages allow humans to give instructions to computers in a structured, readable way
- All code must eventually be translated into machine code (binary) for the CPU to execute
- Programming languages follow different paradigms — procedural, object-oriented, functional, and scripting
- Languages can be low-level or high-level
- Languages are executed either through compilation or interpretation
- Python is a high-level, interpreted, multi-paradigm language, inspired by ABC, created by Guido van Rossum in 1991
- Python is popular due to its readability, versatility, large ecosystem, and strong community support
- Python code runs through the CPython interpreter, converting source code into bytecode executed by the Python Virtual Machine
- You installed Python, set up an editor, and successfully ran your first program

Hands-on Setup



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- Install Python on your computer and verify the installation using `python --version`
- Install a code editor (VS Code, Thonny, or Jupyter Notebook)
- Create a file named `intro.py` and write a program that prints:
 - Your name
 - Today's date
 - One sentence about why you want to learn Python
- Run the program from your terminal using `python intro.py`
- Reflection question (write your answer as a comment in the same file):
 - Is Python a compiled or interpreted language? Explain in one sentence, in your own words.



Next Lecture

- Operators, Expressions & I/O